

DEMO SCRIPT

SolanaAI Lend

3-Minute Video Demo Script

AI-Powered Autonomous Lending on Solana

Gemini AI

6 ML Models

Guard Rails

Live Devnet

Live: <http://89.207.255.254>

National Solana Hackathon by Decentrathon 2026
github.com/Isimbek201632838/solana-service

0:00 — 0:15

Opening

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SolanaAI Lend — an autonomous AI-powered lending protocol on Solana. AI manages interest rates, collateral ratios, and borrow limits in real time, while the smart contract enforces strict safety guardrails. Let me show you how it works.

0:15 — 0:45

Dashboard Overview

SHOW

Interest rate, total deposits, total borrows, SOL price, utilization %

SAY

"Here's our live dashboard. The protocol is running on Solana devnet right now."

POINT

Protocol Mood badge: "This mood indicator — Calm, Cautious, Stressed, or Crisis — reflects the AI's real-time risk assessment."

POINT

Insurance Fund: "10% of all interest flows into this insurance fund to cover bad debt — just like production protocols."

SHOW

Health Factor bar, Liquidation Queue widget

SAY

"Users can see their position health and liquidation risk at a glance."

0:45 — 1:20

AI Decisions Page

CLICK

"AI Decisions" in navigation

SHOW

A decision card with full details

SAY

"Every 2 minutes, the AI agent runs a complete analysis pipeline: 5 technical indicators — RSI, MACD, Bollinger Bands, ATR, EMA — plus 6 ML models for trend prediction, anomaly detection, volatility regime, crash probability, risk scoring, and utilization forecasting."

POINT

Reasoning text: "Gemini 2.0 Flash generates a human-readable explanation for every decision — in English and Russian."

SHOW

Model Performance widget: "The AI tracks each model's accuracy over time and dynamically adjusts their weights. A model that's been wrong loses influence; one that's been right gains it. Self-improving AI."

CLICK

"View TX" → Solana Explorer: "Every parameter change is an on-chain transaction — fully verifiable."

"What makes this different from 'trust the AI' is our 7-layer safety system:"

1 Prompt

Prompt engineering constrains Gemini's output format.

2 Validator

A validator checks 11 rules — rate bounds, collateral bounds, max change per update.

3 Contract

The smart contract enforces the same limits on-chain — impossible to bypass.

4 AI Freeze

AI can emergency-freeze the protocol if crash probability exceeds 80%.

5 Auto-Rate

If the AI agent dies, the contract has an auto-rate mechanism to protect liquidity.

6 Insurance

Insurance fund covers bad debt from failed liquidations.

7 Guardrail PDA

Guardrail parameters are stored in a separate PDA — only authority can modify.

Show log: "CooldownActive — the contract rejected an update because the cooldown hadn't passed. The AI doesn't get to override safety."

Dynamic LTV

VOLATILITY-AWARE

"AI automatically adjusts collateral requirements based on volatility. In calm markets — lower collateral to attract borrowers. In a storm — collateral goes up 10-20% to protect the pool."

→ Show LTV Mode badge (Calm / Normal / Storm / Extreme)

Credit Score

ON-CHAIN IDENTITY

"Every wallet gets an on-chain credit score based on 5 factors: account age, transaction count, repayment history, liquidation record, and position size."

→ Show Credit Score widget: "Platinum users get 15% lower collateral and 10% lower rates."

Crash Detection

6 SIGNALS

"Six signals — price momentum, volume spikes, consecutive red candles, SMA deviation, short-term volatility, and trend acceleration — feed into a crash probability score. Above 80%? The AI freezes the protocol automatically."

Liquidation Predictor

MONTE CARLO

"Monte Carlo simulation with 500 price paths predicts liquidation probability at 1-hour, 4-hour, and 24-hour horizons. Users get warnings before they're in danger."

2:30 — 2:50

Architecture & Resilience

Nginx

Reverse proxy

React

Frontend dashboard

FastAPI

Backend + WebSocket

AI Agent

ML pipeline + Gemini

Simulator

Activity generator

Resilience

If Gemini goes down → ML-only fallback decisions.

If the entire agent dies → contract's auto-rate kicks in.

The protocol NEVER stops.

2:50 — 3:00

Closing

"SolanaAI Lend: AI thinks, the contract controls."

"Not 'trust the AI' — but 'verify on-chain'."

Live: <http://89.207.255.254>

github.com/IsIambek201632838/solana-service

Recommended Screenshots

1

Dashboard with live data (desktop)

2

AI Decision card with ML metrics + reasoning

3

Solana Explorer — TX proof

4

Credit Score widget showing tier

5

Liquidation Warning with Monte Carlo

6

LTV Mode badge changing with volatility

7

Mobile responsive view

8

Architecture diagram